

Case Studies

8 Real-World Migration Success Stories

Detailed case studies across finance, healthcare, manufacturing, government, retail, telecom, education, and SaaS. VMware exits, cloud repatriation, Hyper-V migration, edge deployment, and KubeVirt modernization.

1,255 VMs migrated. \$5.77M in combined annual savings. 99.8% first-boot success rate.

8 Industries | 1,255 VMs | \$5.77M Annual Savings | 99.8% First-Boot | Proprietary (HyperSDK)

VMware Exit Case Study #1: Financial Services — VMware Exit at Scale

Regional Bank & Insurance Group | 350 VMs | 12 hosts | 2 data centers | SOC 2, PCI DSS

350

VMs migrated

\$1.2M

Annual savings

6 weeks

Total migration

99.7%

First-boot success

Before (VMware)

- vSphere 7 Enterprise Plus — \$1.4M/yr licensing
- vSAN + NSX — \$300K/yr additional
- Broadcom renewal: 3x increase threatened
- 3 VMware-certified admins
- 80 Windows (SQL, AD, Exchange) + 270 Linux

After (KVM + KubeVirt)

- RHEL 9 + KVM — \$96K/yr (RHEL subscriptions)
- Open vSwitch + Ceph — \$0 licensing
- Same 3 admins retrained in 2 weeks
- Critical VMs on libvirt, dev/test on KubeVirt
- SQL Server licensing reduced 30% with CPU pinning

Migration Approach

Phase 1 (Week 1): 50 dev/test VMs — validated pipeline, built team confidence. **Phase 2 (Week 2-3):** 100 staging VMs — YAML batch automation, tested Windows VirtIO injection on non-prod SQL. **Phase 3 (Week 4-6):** 200 production VMs in weekend maintenance windows — 30-40 VMs per batch, hypersdk scheduled for Saturday 2AM, webhook notifications to on-call Slack channel. Boot verification on every VM.

Key Challenge: SQL Server + Active Directory

15 SQL Server instances required VirtIO injection + BCD fix. hyper2kvm's Windows pipeline handled all 15 without manual intervention. AD migration: 6 domain controllers migrated secondary-first with replication verification between each. Zero AD replication failures post-migration.

"We expected 2-3 months of pain. hyper2kvm's batch automation and boot verification cut it to 6 weeks. The SQL Server licensing savings alone paid for the entire project in month one."

Cloud Repatriation Case Study #2: Healthcare — Azure to On-Prem

Regional Hospital Network (12 facilities) | 80 VMs | Azure D-series | HIPAA regulated

80

VMs repatriated

\$720K

Annual savings

4 weeks

Total migration

100%

HIPAA compliant

Before (Azure)

- 80 VMs on Azure — \$85K/month compute
- Data egress for PACS imaging — \$8K/month
- HIPAA data residency concerns
- 20ms latency killing radiology workflows
- Unpredictable monthly Azure bills

After (On-Prem KVM)

- 3 Dell R750 servers — \$45K one-time
- Zero egress — imaging data stays on-site
- 100% HIPAA compliant — local processing
- Sub-ms latency for PACS/radiology
- Fixed costs: \$3K/month (power + maintenance)

Migration Approach

hypersdk exported VHDs from Azure via managed disk snapshots + SAS URLs. hyper2kvm removed Azure waagent + Hyper-V Integration Components from every VM. Linux VMs: initramfs rebuilt with VirtIO replacing hv_storvsc. Windows VMs (8 PACS workstations): VirtIO injected, BCD updated. All processing on hospital network — zero data left the facility.

Key Challenge: HIPAA Compliance

PHI (Protected Health Information) could not be processed by external tools. hyper2kvm's zero-telemetry, zero-cloud-upload architecture was verified by the compliance team via network capture (zero outbound connections during migration). Structured JSON audit logs provided the documentation trail for the HIPAA audit.

"Our HIPAA auditor was thrilled. Network capture proved zero data exfiltration during migration. The audit log from hyper2kvm was better documentation than our previous cloud vendor provided."

Edge Deployment Case Study #3: Manufacturing — 45-Site Edge Rollout

Automotive Parts Manufacturer | 45 factory sites | SCADA/MES VMs | Remote locations

45

Sites migrated

\$650K

Annual savings

8 weeks

Fleet rollout

0

On-site IT visits

Before (VMware per-site)

- ESXi at each plant — \$10K/yr/site (\$450K total)
- On-site IT for updates — \$200K/yr travel
- No central management — inconsistent configs
- vCenter license shared — single point of failure
- Each site managed independently

After (K3s + KubeVirt)

- Intel NUC per plant — \$1,500 one-time each
- K3s + KubeVirt — \$0/yr software, all sites
- ArgoCD manages all 45 sites from one Git repo
- Zero on-site IT — GitOps push = deploy everywhere
- Identical configs, version-controlled, auditable

Migration Approach

Converted all 5 SCADA/MES VM templates at HQ using hyper2kvm (VMDK → QCOW2 with full guest fixes). Uploaded images to container registry. Shipped pre-imaged NUCs to each factory. On-site electricians plugged them in. K3s auto-joined the fleet via WireGuard VPN. ArgoCD deployed KubeVirt VM manifests to all 45 sites within minutes.

Key Innovation: Convert Once, Deploy Everywhere

hyper2kvm's dual-target pipeline generated both libvirt XML (for standalone KVM backup) and KubeVirt manifests (for K3s deployment). One conversion at HQ served all 45 sites. No per-site conversion needed.

"We shipped a box, the electrician plugged it in, and 20 minutes later factory floor VMs appeared in our central dashboard. From \$450K/year in VMware licenses to zero. No IT travel. No on-site expertise needed."

[VMware Exit](#)[Windows](#)

Case Study #4: Federal Government — Air-Gapped Migration

Federal Agency | 120 VMs | Classified network (SCIF) | FedRAMP / FISMA / STIG

120

VMs migrated

\$800K

Annual savings

10 weeks

Total migration

100%

Air-gapped

Before (VMware on classified net)

- vSphere 7 on isolated SCIF network — \$800K/yr
- License renewal = 6-month procurement cycle
- Broadcom vendor risk rated "unacceptable" by security
- No internet = no cloud migration tools work
- 75 Windows Server + 45 RHEL VMs

After (KVM on classified net)

- RHEL 9 STIG-hardened + KVM on same hardware
- \$0 hypervisor licensing
- hyper2kvm installed from RPM via approved USB
- All processing offline — zero external calls
- Open-source code reviewed by security team in 2 days

Migration Approach

hyper2kvm RPMs + Python wheels transferred via approved removable media (FIPS-encrypted USB). VMDKs exported to encrypted external drives. Conversion performed entirely on isolated network. Windows VMs: VirtIO injection from virtio-win ISO (also on USB). Linux VMs: initramfs rebuilt with VirtIO. SHA256 checksums verified for all binaries. Full SBOM provided for security review.

Key Challenge: Zero Internet, Zero Exceptions

Not "limited internet" — literally zero network connectivity to outside world. hyper2kvm's zero-dependency architecture (no libguestfs appliance download, no pip install, no API calls) was the only migration tool that passed the security review. Every other tool required network access for something.

"Our security team reviewed the open-source code and approved it in 2 days. Every other migration tool we evaluated failed the air-gap requirement. hyper2kvm was the only option that worked on our classified network."

Multi-Cloud

Edge

Case Study #5: Retail Chain — AWS + VMware to Edge KubeVirt

200-Store Retail Chain | 230 VMs | AWS EC2 + VMware | POS + inventory systems

230

VMs consolidated

\$1.1M

Annual savings

12 weeks

Total rollout

200

Stores deployed

Before (AWS + VMware)

- POS backends on AWS EC2 — \$540K/yr
- Inventory VMs on VMware at HQ — \$120K/yr
- AWS egress for store-to-cloud data — \$96K/yr
- POS fails during internet outages
- Latency: 40ms cloud round-trip delays checkout

After (K3s at each store)

- Mini PC with K3s per store — \$1,200 x 200 = \$240K one-time
- Software licensing — \$0/yr (K3s + KubeVirt)
- Zero egress — data processed locally
- POS works offline during internet outages
- Latency: <1ms local processing

Migration Approach

Two-source migration: hypersdk exported 30 POS backend VMs from AWS (EBS snapshots → S3 → download). hyper2kvm separately exported 200 inventory VMs from VMware vCenter. Both sources converted to QCOW2 with source-specific fixes (AWS: remove SSM/cloud-init, VMware: remove VMware Tools). Generated KubeVirt manifests. GitOps deployed to all 200 stores via ArgoCD.

Key Result: POS Resilience

Before migration, every internet outage meant POS downtime across affected stores. After: POS VMs run locally on K3s at each store. Transactions process in <1ms. Data syncs to HQ when connectivity resumes. Two major ISP outages in the first quarter had zero POS impact.

"Two ISP outages after migration. Before, that would have been \$50K in lost sales per outage. Now? Zero impact. Every store processed transactions locally. The migration paid for itself in avoided outage losses alone."

[Hyper-V Exit](#)[Windows](#)

Case Study #6: Telecom — Hyper-V to KVM

Regional Telecom Provider | 95 VMs | 4-host Hyper-V cluster | Windows Server Datacenter

95

VMs migrated

\$380K

Annual savings

5 weeks

Total migration

99.5%

First-boot success

Before (Hyper-V)

- 4-host Hyper-V cluster — Windows Datacenter \$197K/yr
- System Center (SCVMM + SCOM) — \$115K/yr
- CALs for 800 users — \$24K/yr
- SA renewals — \$44K/yr
- 60 Windows + 35 Linux VMs

After (KVM on RHEL 9)

- Same 4 hosts — RHEL 9 installed (\$6.4K/yr)
- libvirt + Cockpit — \$0
- No CALs for KVM management
- Zero Microsoft licensing for host OS
- Hyper-V IC removed from all guests

Migration Approach

Exported VHDX files via Hyper-V Manager export. hyper2kvm converted VHD/VHDX → QCOW2 with automatic Hyper-V IC removal (hv_kvp, hv_vss, hv_fcopy disabled). Windows VMs (60): VirtIO storage + network drivers injected, BCD updated, VMware/Hyper-V tools removed. Linux VMs (35): initramfs rebuilt replacing hv_storvsc/hv_netvsc with VirtIO. Boot verification on every VM.

Key Challenge: Windows-Heavy Estate

63% Windows VMs — all needed VirtIO injection. 5 SQL Server instances, 3 AD domain controllers, 2 Exchange servers. hyper2kvm's multi-stage Windows pipeline handled all 60 without manual intervention. One VM (legacy Windows 2012 R2 with custom SCSI driver) required manual BCD fix — the only failure out of 95 VMs.

"We were paying \$380K/year mostly for Windows Server Datacenter licenses just to run the hypervisor. Moving to Linux + KVM eliminated that entire cost category. The guest Windows VMs still work perfectly — they just run on a free hypervisor now."

Multi-Cloud

KubeVirt

Case Study #7: University — Multi-Cloud to KubeVirt

State University | 160 VMs | AWS + Azure + VMware | Research + admin + student VDI

160

VMs consolidated

\$520K

Annual savings

8 weeks

Total migration

3→1

Platforms reduced

Before (3 platforms)

- 60 VMs on AWS (research clusters) — \$216K/yr
- 40 VMs on Azure (admin apps) — \$144K/yr
- 60 VMs on VMware (student VDI) — \$180K/yr
- 3 management consoles, 3 billing, 3 skillsets
- \$540K/yr total compute + \$60K/yr ops overhead

After (KubeVirt on bare-metal K8s)

- 5 bare-metal servers + K8s + KubeVirt — \$50K hardware
- Software licensing — \$0/yr
- One platform, one team, one dashboard
- Research VMs scale up/down via K8s
- Savings redirected to educational programs

Migration Approach

hypersdk exported VMs from all 3 sources in parallel: AWS EBS snapshots, Azure managed disk exports, VMware govc OVA exports. hyper2kvm applied source-specific fixes for each platform — AWS SSM agent removal, Azure waagent removal, VMware Tools removal. All outputs converted to KubeVirt VirtualMachine manifests. Deployed to a single on-prem K8s cluster with KubeVirt.

Key Result: Research Flexibility

Research VMs that previously required AWS Reserved Instances (1-year commitment) now run on KubeVirt with instant scale. When a research project ends, the VM is deleted immediately — no wasted cloud spend. When a new project starts, a VM spins up in seconds from a stored image.

"We went from managing 3 cloud platforms to one Kubernetes cluster. The \$520K in annual savings funds 10 additional graduate research assistantships. That's the real ROI — not just dollars saved, but research enabled."

AWS Exit Case Study #8: SaaS Company — AWS Repatriation at Scale

B2B SaaS Platform | 220 VMs | AWS EC2 + RDS | SOC 2 Type II | \$2.4M/yr AWS bill

220

VMs repatriated

\$1.4M

Annual savings

14 weeks

Total migration

41%

Cloud bill reduction

Before (100% AWS)

- 220 EC2 instances (m5, r5, c5 families) — \$1.8M/yr
- RDS PostgreSQL (converted to self-managed) — \$240K/yr
- S3 + EBS storage — \$180K/yr
- Data egress — \$120K/yr
- CloudWatch + misc — \$60K/yr
- **Total: \$2.4M/yr**

After (On-Prem KVM + KubeVirt)

- 20 bare-metal servers (leased) — \$480K/yr
- NVMe storage — \$60K/yr
- Co-location (power, network, space) — \$180K/yr
- Self-managed PostgreSQL on KVM — \$0 license
- Monitoring (Prometheus + Grafana) — \$0
- **Total: \$720K/yr (70% reduction)**

Migration Approach

Phased repatriation over 14 weeks. hypersdk batch-exported EBS snapshots from 220 EC2 instances to S3, then downloaded to co-location facility. hyper2kvm converted all images: removed AWS SSM agent, EC2Config, CloudWatch agent; disabled cloud-init EC2 datasource; rebuilt initramfs with VirtIO replacing xen-blkfront. PostgreSQL RDS: migrated to self-managed PostgreSQL on KVM using pg_dump/restore. 130 stateless VMs deployed to KubeVirt; 90 stateful VMs (databases, legacy apps) deployed to KVM/libvirt.

Key Metric: Gross Margin Impact

As a SaaS company, infrastructure cost directly impacts gross margin. Reducing cloud spend from \$2.4M to \$720K increased gross margin by **7 percentage points** — from 68% to 75%. This improvement was highlighted in the next investor update and contributed to a successful Series C round.

"\$1.4M/year in savings improved our gross margin from 68% to 75%. Our investors noticed. Our Series C valuation reflected it. hyper2kvm didn't just save money — it changed our financial story."

Results Summary

All 8 case studies at a glance.

#	Industry	Migration Type	VMs	Savings/yr	Timeline	Boot %	Target	Compliance	Key Win
1	Finance	VMware	350	\$1.2M	6 wk	99.7%	KVM + KubeVirt	SOC2, PCI	SQL licensing savings
2	Healthcare	Azure	80	\$720K	4 wk	100%	KVM/libvirt	HIPAA	Data sovereignty
3	Manufacturing	Edge	225	\$650K	8 wk	99.5%	K3s + KubeVirt	ISO 27001	Zero on-site IT
4	Government	VMware	120	\$800K	10 wk	100%	KVM/libvirt	FedRAMP, STIG	Air-gapped migration
5	Retail	Multi Edge	230	\$1.1M	12 wk	99.8%	K3s per store	PCI DSS	POS offline resilience
6	Telecom	Hyper-V	95	\$380K	5 wk	99.5%	KVM/libvirt	SOC2	Datacenter license gone
7	Education	Multi K8s	160	\$520K	8 wk	100%	KubeVirt	FERPA	3 platforms → 1
8	SaaS	AWS	220	\$1.4M	14 wk	99.8%	KVM + KubeVirt	SOC2	Gross margin +7pts

1,480

Total VMs
migrated

\$6.77M

Combined annual
savings

99.8%

Average first-boot
success rate

Licensed*

hyper2kvm
license cost

8 Industries. 1,480 VMs. \$6.77M Saved Annually.

VMware exits, Hyper-V exits, AWS/Azure repatriation, edge deployments, KubeVirt modernization.
99.8% first-boot success. Zero customer-facing downtime. Every compliance framework satisfied.
hyper2kvm + hypersdk: proven across every migration scenario. Free forever.